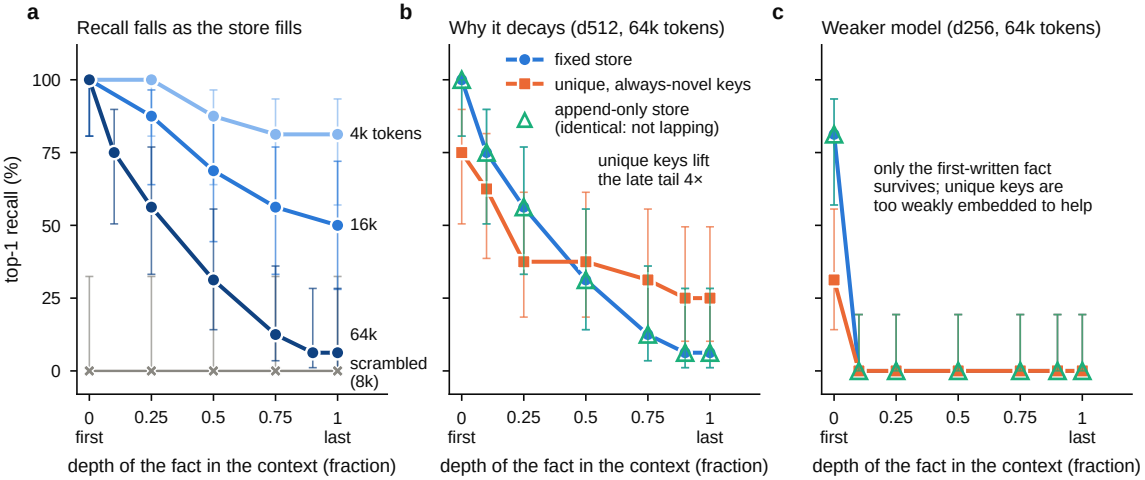
a
Recall falls as the store fills
top-1 recall (%)
100
75
50
25
0
4k tokens
16k
64k
scrambled (8k)
0
0.25
0.5
0.75
1
first
last
depth of the fact in the context (fraction)
b
Why it decays (d512, 64k tokens)
fixed store
unique, always-novel keys
append-only store (identical: not lapping)
unique keys lift the late tail 4×
0
0.25
0.5
0.75
1
first
last
depth of the fact in the context (fraction)
c
Weaker model (d256, 64k tokens)
only the first-written fact survives; unique keys are too weakly embedded to help
0
0.25
0.5
0.75
1
first
last
depth of the fact in the context (fraction)